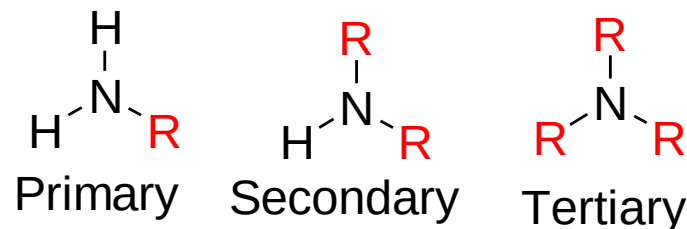


Amines

CHE 1203 – Organic Chemistry 1

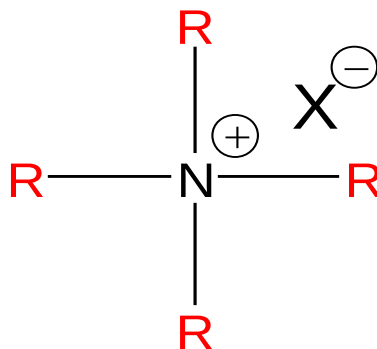
Naming Amines

- Alkyl-substituted (alkylamines) or aryl-substituted (arylamines)
- Classified:
 - 1° (RNH_2) primary amine
 - 2° (R_2NH) secondary amine
 - Tertiary amine 3° (R_3N)



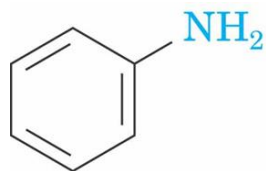
Quaternary Ammonium Ions

- A nitrogen atom with four attached groups is positively charged
- Compounds are quaternary ammonium salts

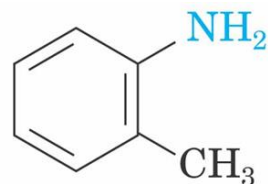


Common Names

- Alkylamines do not have common names unless they are biological molecules
- Simple arylamines have common names



Aniline



***o*-Toluidine**

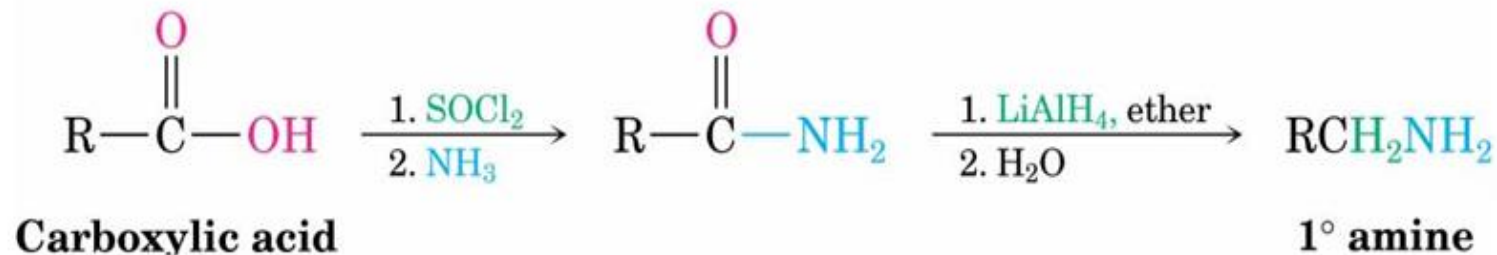
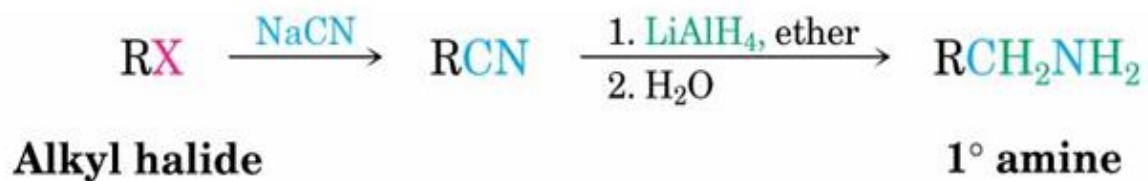
Properties and Sources of Amines

- Simple methylated amines from reaction of NH_3 with CH_3OH and alumina catalyst
- Yields a mixture of monomethylated, dimethylated, and trimethylated products that are easily separated by distillation



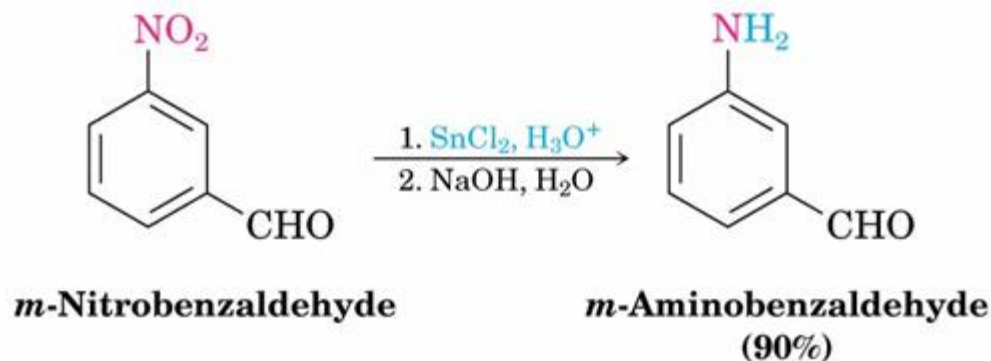
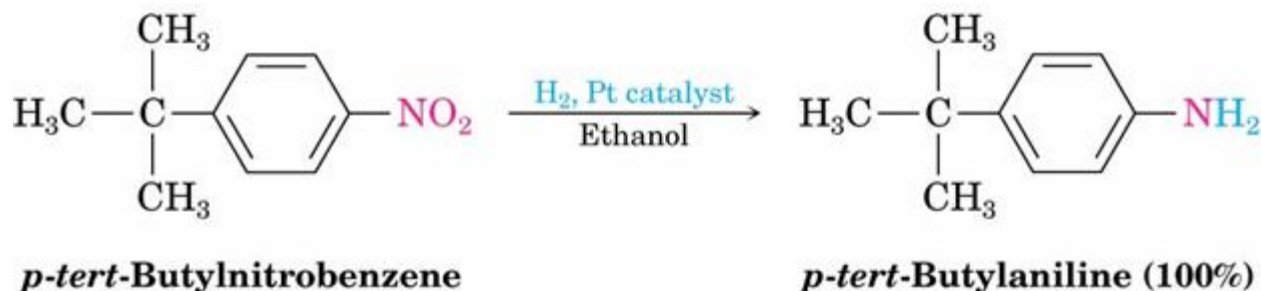
Synthesis of Amines

- Reduction of nitriles and amides



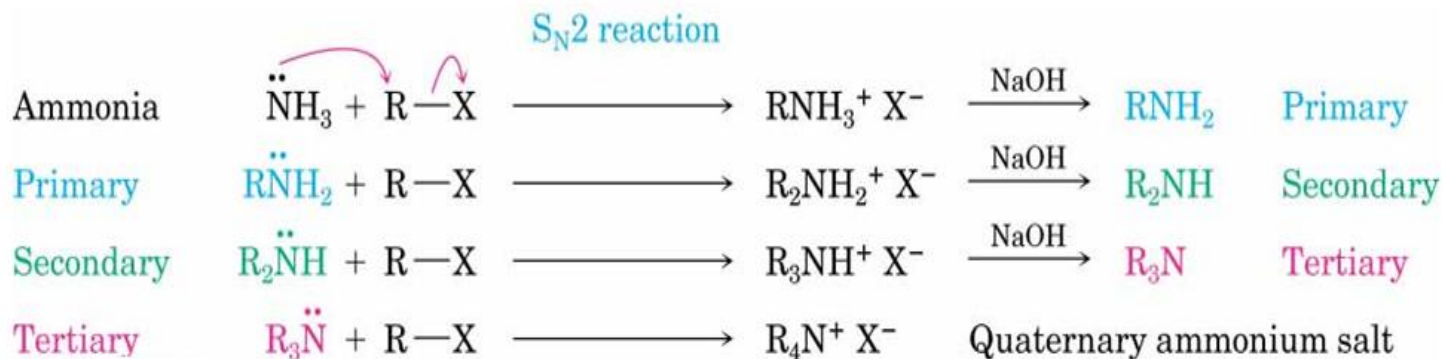
Reduction Aryl Nitro Compounds

- Arylamines are prepared from nitration of an aromatic compound and reduction of the nitro group
- Reduction by catalytic hydrogenation over platinum is suitable if no other groups can be reduced
- Iron, zinc, tin, and tin(II) chloride are effective in acidic solution



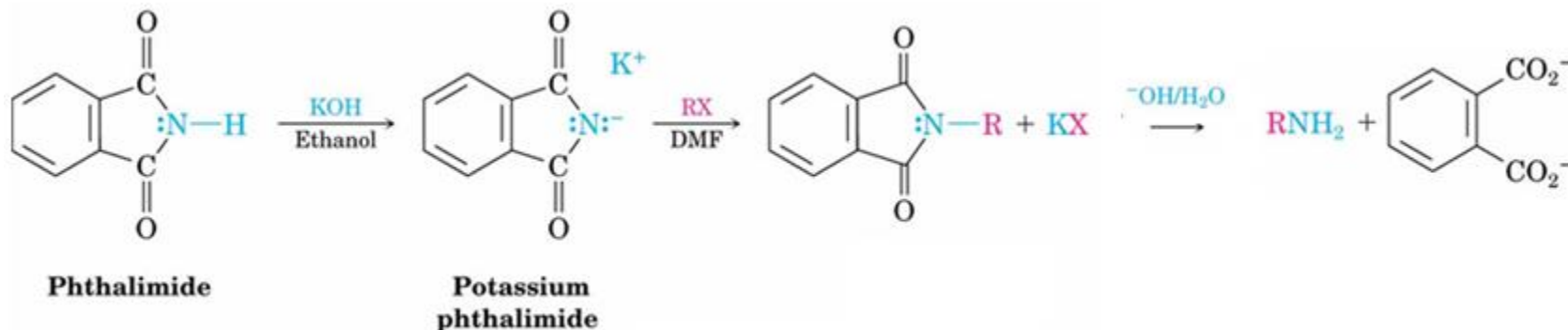
Reactions of Alkyl Halides

- Ammonia and other amines are good nucleophiles



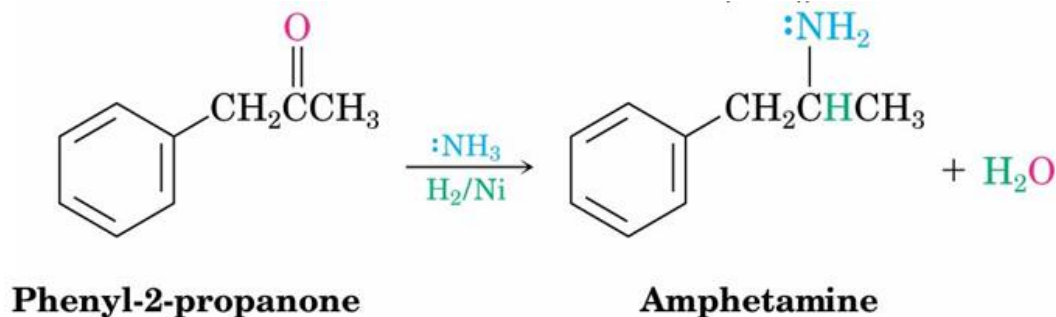
Gabriel Synthesis of Primary Amines

- A *phthalimide* alkylation for preparing a primary amine from an alkyl halide
- The N-H in imides (—CONHCO—) can be removed by KOH followed by alkylation and hydrolysis



Reductive Amination of Aldehydes and Ketones

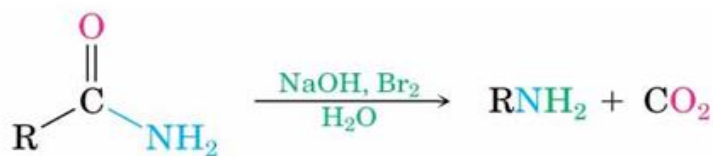
- Treatment of an aldehyde or ketone with ammonia or an amine in the presence of a reducing agent



Hofmann and Curtius Rearrangements

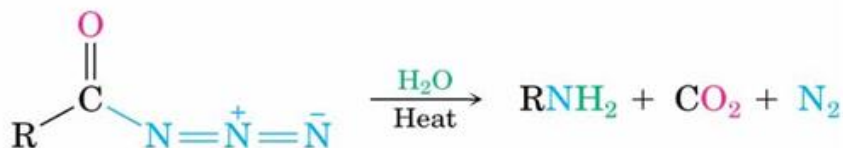
- Carboxylic acid derivatives can be converted into primary amines with loss of one carbon atom by both the Hofmann rearrangement and the Curtius rearrangement

**Hofmann
rearrangement**



An amide

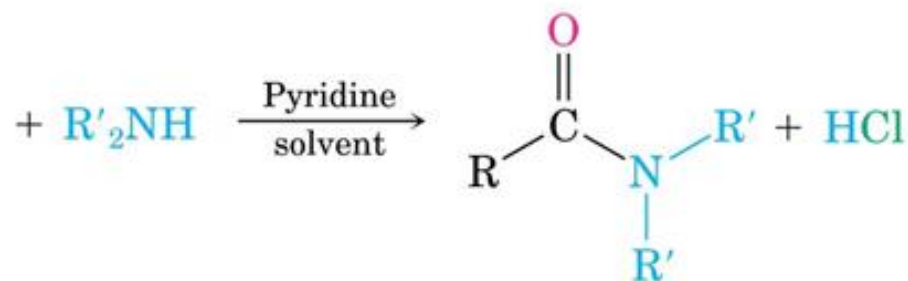
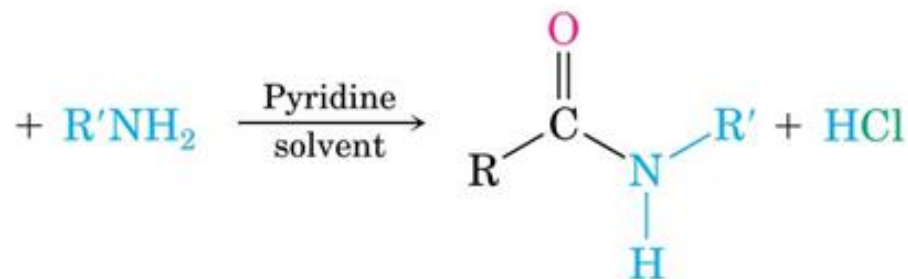
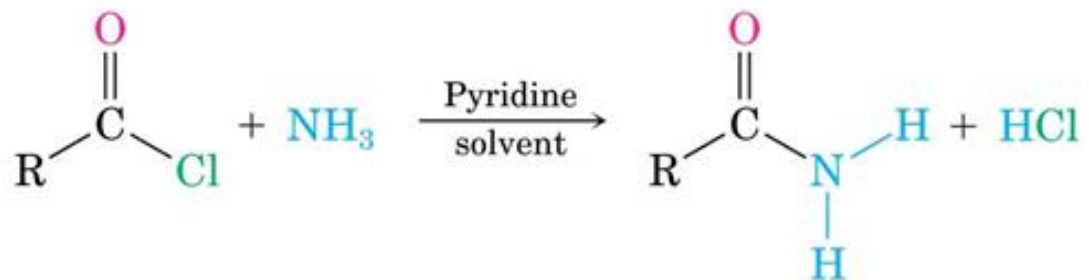
**Curtius
rearrangement**



An acyl azide

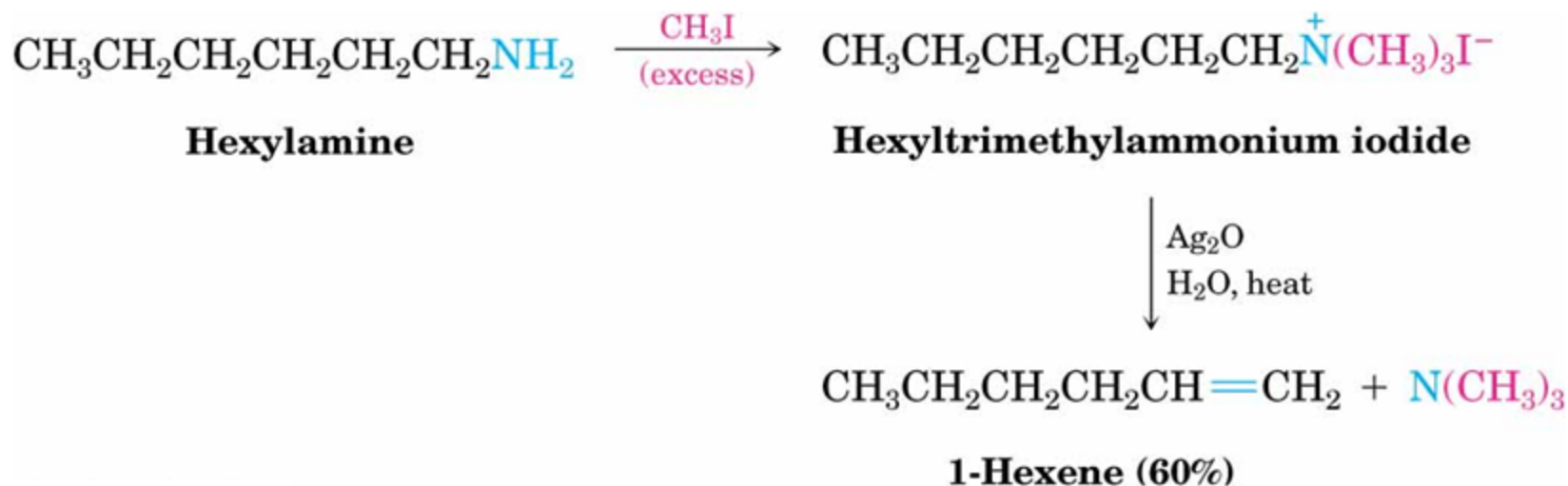
Reactions of Amines

- Alkylation and acylation have already been presented



Hofmann Elimination

- Converts amines into alkenes
- NH_2^- is very a poor leaving group so it converted to an alkylammonium ion, which is a good leaving group



Selective Preparation of Primary Amines: the Azide Synthesis

- Azide ion, N_3^- displaces a halide ion from a primary or secondary alkyl halide to give an alkyl azide, RN_3
- Alkyl azides are not nucleophilic (but they are explosive)
- Reduction gives the primary amine

